

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

February 2023

Course Code: CHY101

Course: ENGINEERING CHEMISTRY

QP No. :U018R-1

Duration: 3 hours

Max. Marks:100

PART-A

Answer all the questions

10 x 2 = 20 Marks

1. During EDTA titration with hard water, M-EBT and M-EDTA complexes will be formed. Compare the stability of these complexes.
2. One mL of EDTA solution is equivalent to 2.0 mg CaCO_3 . 50 mL of the sample hard water was titrated with EDTA solution and it consumed 15.0 mL EDTA solution. Calculate total hardness of water sample.
3. At the end of the disinfection process, the hardness of water was found to increase. Identify the disinfectant used and provide the reason for the increase in hardness.
4. Predict whether the given reaction is spontaneous or not:
 $\text{Fe} + \text{Pb}^{2+} \rightarrow \text{Fe}^{2+} + \text{Pb}$; $E_{\text{Fe}^{2+}/\text{Fe}}^0 = -0.44 \text{ V}$; $E_{\text{Pb}^{2+}/\text{Pb}}^0 = -0.13 \text{ V}$;
5. For food packaging, which one of the following is preferred & justify your answer.
(a) Stainless steel container; (b) Tin container; (c) Galvanized iron container.
6. For a particular type of corrosion, rate of corrosion decreases with increase in temperature. Identify the corrosion and justify your answer.
7. The viscosity index of the following lubricants are tabulated below:

Entry	Lubricant	V.I
A	Mineral Oil	95-105
B	Silicone oil	205-400
C	Polyglycols	140-200

Identify the lubricant that can be employed for machines which experiences increase in temperature during operation.

8. Although the molecular weight of ethanol and dimethyl ether are same, dimethyl ether is more volatile than ethanol. Give reason.
9. "Gold in bulk form is unreactive, but gold nanoparticles can be used as catalyst". Justify the above statement.
10. Identify the matrix and dispersed phase of concrete and classify under which type of composite it belongs.

PART-B

Answer all the questions

4 x 15 = 60 Marks

11. (a) 0.5 g of CaCO_3 was dissolved in HCl and it is made up to 1 L with distilled water. 50 mL of this solution required 42 mL of EDTA. 50 mL of sample water taken from corporation water treatment plant required 25 mL of EDTA. The sample water was then boiled and filtered and the 50 mL of the filtrate required 15 mL of EDTA. Calculate various types of hardness and express it in ppm or CaCO_3 equiv.
- (b) The same sample water without heating was further softened by zeolite-based softener. To reactivate the softener, 250 L of NaCl solution containing 100 g/L was employed. How many liters of the sample water can be purified by the zeolite process?

(OR)

12. (a) For purifying water for boiler industries, AquaTech & Co. employs different internal treatment methods. With suitable equations, suggest suitable internal treatment employed for water having following characteristics:
- (i) Water with high level of Calcium salt.
 - (ii) Water obtained from lake contaminated with acidic effluent from neighboring industries
- b) "Direct current can be used for treatment of brackish water along with suitable membranes". Explain the working principle for such process. How will you improve the efficiency for such a process?

13. (a) Qualitative analysis of a water sample shows that the water is contaminated with cadmium. To calculate the concentration of Cd^{2+} ions contamination in the water sample, an analyst constructed a galvanic cell with a Cd rod dipped in water containing Cd^{2+} ions of unknown concentration and Ag wire dipped in AgNO_3 solution (0.245 M). Write the cell representation and half-cell reactions. The measured EMF of the cell is 1.23 V at 25 °C. $E^\circ(\text{Cd}^{2+}/\text{Cd}) = -0.40$ V and $E^\circ(\text{Ag}^+/\text{Ag}) = +0.80$ V respectively. Write the half-cell reactions and calculate the concentration of Cd^{2+} ion present in water.
- (b) The Cd^{2+} ions might have leached from the trash of a used secondary battery. Identify the secondary battery and write the working principle of the battery by providing discharging and charging equations.

(OR)

14. The main building material of ships are either iron/steel, thus it's impossible to prevent corrosion during sailing.
- (a) Identify the different types of corrosion that can take place in the ship during voyage in high sea
- (b) Suggest any two suitable inorganic coatings in detail to prevent corrosion.

15. Classify the given compounds which *can* and *cannot* form H-bond with themselves. If the compound can form H-bond with itself, arrange them in increasing order of hydrogen bond strength. Justify your answer for both the cases.

(a) $(\text{CH}_3)_3\text{N}$ (b) C_6H_6

(c) HF

(d) CHCl_3

(e) Water (f)

NH_3

(OR)

16. (a) The test oil has same viscosity as Gulf oil and Pennsylvanian oil at 210 °F. The viscosities at 100 °F are 350 s, 420 s and 280 s for test oil, Gulf oil and Pennsylvanian oil respectively. Calculate the Viscosity Index (VI) of test oil. Also, based on the data draw a graph which shows the variation of viscosity with temperature.
- (b) 50 mL of mineral oil is accidentally spilled into a container having 1.0 L of high-grade olive oil used for cooking. As an analyst who is in-charge of grading the oil, how will you identify the presence of mineral oil contamination in high-grade olive oil?

17. Identify the classification of the following composite materials. List out the properties of these composite materials.

- (a) Bituminous road
- (b) Plywood
- (c) Cardboard sheet

(OR)

18. A pharmaceutical company want to develop a drug based on nanoparticle that can destroy cancerous cell more selectively than normal cells. The inventory has various Ag, Zn, Au and Ti salts.

- (a) Choose the appropriate salt that can be employed for the aforementioned application and discuss the synthetic methodology for the preparation of this material.
- (b) Give a note on of anti-cancer activity exhibited by the synthesized nanoparticle.

PART-C

Answer the following

1 x 20 = 20 Marks

19. The prime reason for life in earth is due to the presence of "*Water*" and it can be rightly known as "elixir of life" and based on these thoughts, give your insights for the following:

- (a) During a rainy day, imagine a droplet of rainwater which falls on the surface of leaf and a dirty glass window. Predict the shape of water on leaf's surface and on dirty glass window? Give reason for your prediction.
- (b) During steam production, presence of trace amount of oil inside the boiler will cause extensive foaming. Give reason for this phenomenon and this can be controlled by another oil – sulfated castor Oil. Decipher the reason behind this contradiction.
- (c) It is feasible to form water from its constituents (i.e) hydrogen and oxygen, but the reverse reaction is not thermodynamically feasible. Justify the statement with proper equation and identify the reverse reaction, which plays an important role for generation of renewable and cleaner fuel – Hydrogen.
- (d) Water purification can be performed using this coinage metal derived nanoparticle. Identify the nanoparticle employed for water purification and elucidate the mechanism of water purification aided by the aforementioned nanoparticles.
